ADI-FDTD for Inhomogeneous Media

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1 Preliminaries

1.1 Maxwell's Equations

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} \tag{1.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \tag{1.1b}$$

Expanding (1.1a) into components:

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \varepsilon \frac{\partial E_x}{\partial t} + \sigma E_x \tag{1.2a}$$

$$\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} = \varepsilon \frac{\partial E_y}{\partial t} + \sigma E_y \tag{1.2b}$$

$$\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = \varepsilon \frac{\partial E_z}{\partial t} + \sigma E_z \tag{1.2c}$$

Expanding (1.1b) into components:

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right) \tag{1.3a}$$

$$\frac{\partial H_y}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} \right) \tag{1.3b}$$

$$\frac{\partial H_z}{\partial t} = \frac{1}{\mu} \left(\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \right) \tag{1.3c}$$

1.2 Yee Grid in AMReX

The Yee grid stores different field components at different physical locations within each computational cell. In AMReX this staggering is represented by the `IndexType` used to define each `MultiFab`: a cell-centered direction has index type 0 and is located at $i + \frac{1}{2}$, while a nodal direction has index type 1 and is located at i. The implementation uses

$$\mathbf{E}_\alpha : \text{cell-centered in the } \alpha \text{ direction and nodal in the other two directions,}$$
$$\mathbf{H}_\alpha : \text{nodal in the } \alpha \text{ direction and cell-centered in the other two directions.}$$

Thus the component locations are

Component	x location	y location	z location
$E_{x,i,j,k}$	$i + \frac{1}{2}$	j	k
$E_{y,i,j,k}$	i	$j + \frac{1}{2}$	k
$E_{z,i,j,k}$	i	j	$k + \frac{1}{2}$
$H_{x,i,j,k}$	i	$j + \frac{1}{2}$	$k + \frac{1}{2}$
$H_{y,i,j,k}$	$i + \frac{1}{2}$	j	$k + \frac{1}{2}$
$H_{z,i,j,k}$	$i + \frac{1}{2}$	$j + \frac{1}{2}$	k

The half-cell offsets therefore do not appear explicitly as array indices such as $i+\frac{1}{2}$. They are encoded in the `MultiFab` staggering. The integer offsets in the finite-difference stencil choose the two source values that bracket the field being updated. For example, $H_{x,i,j,k}$ is located at $j+\frac{1}{2}$ in the y direction, while $E_{z,i,j,k}$ and $E_{z,i,j+1,k}$ are located at j and $j+1$. Therefore

$$\left.\frac{\partial E_z}{\partial y}\right|_{H_{x,i,j,k}} \approx \frac{E_{z,i,j+1,k} - E_{z,i,j,k}}{\Delta y}. \tag{1.4}$$

Similarly, $E_{x,i,j,k}$ is located at j, while $H_{z,i,j-1,k}$ and $H_{z,i,j,k}$ are located at $j-\frac{1}{2}$ and $j+\frac{1}{2}$, so

$$\left.\frac{\partial H_z}{\partial y}\right|_{E_{x,i,j,k}} \approx \frac{H_{z,i,j,k} - H_{z,i,j-1,k}}{\Delta y}. \tag{1.5}$$

This is why the magnetic update below uses forward differences such as $j+1$ and $k+1$, while the electric update uses backward differences such as $j-1$ and $k-1$: each derivative is centered at the staggered location of the field component being updated.

2 FDTD Method

The explicit Yee FDTD update used here is written in three stages, matching the implementation: first update $\mathbf{H}$ by a half step, then update $\mathbf{E}$ by a full step, and finally update $\mathbf{H}$ by another half step. Define the update coefficients

$$C_a = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{\mu}, \tag{2.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity.

2.1 First Magnetic Half-Step

$$H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^n_{z,i,j+1,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^n_{y,i,j+\frac{1}{2},k+1} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.2a}$$

$$H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^n_{z,i+1,j,k+\frac{1}{2}} - E^n_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.2b}$$

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^n_{y,i+1,j+\frac{1}{2},k} - E^n_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^n_{x,i+\frac{1}{2},j+1,k} - E^n_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.2c}$$

In the AMReX Yee grid, with vacuum media, the magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k) - E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1) - E_y(i,j,k)}{\Delta z}\right] \tag{2.3a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1) - E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}\right] \tag{2.3b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k) - E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k) - E_x(i,j,k)}{\Delta y}\right]. \tag{2.3c}$$

2.2 Electric Full-Step

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C^{n+1}_{a,i+\frac{1}{2},j,k} E^n_{x,i+\frac{1}{2},j,k} + C^{n+1}_{b,i+\frac{1}{2},j,k}\left[\frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}\right] \tag{2.4a}$$

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C^{n+1}_{a,i,j+\frac{1}{2},k} E^n_{y,i,j+\frac{1}{2},k} + C^{n+1}_{b,i,j+\frac{1}{2},k}\left[\frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}\right] \tag{2.4b}$$

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C^{n+1}_{a,i,j,k+\frac{1}{2}} E^n_{z,i,j,k+\frac{1}{2}} + C^{n+1}_{b,i,j,k+\frac{1}{2}}\left[\frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}\right]. \tag{2.4c}$$

In the AMReX Yee grid, with vacuum media, the electric field updates are implemented as

$$E_x(i,j,k) = E_x(i,j,k) + c^2\Delta t\left[\frac{B_z(i,j,k)-B_z(i,j-1,k)}{\Delta y} - \frac{B_y(i,j,k)-B_y(i,j,k-1)}{\Delta z}\right] \tag{2.5a}$$

$$E_y(i,j,k) = E_y(i,j,k) + c^2\Delta t\left[\frac{B_x(i,j,k)-B_x(i,j,k-1)}{\Delta z} - \frac{B_z(i,j,k)-B_z(i-1,j,k)}{\Delta x}\right] \tag{2.5b}$$

$$E_z(i,j,k) = E_z(i,j,k) + c^2\Delta t\left[\frac{B_y(i,j,k)-B_y(i-1,j,k)}{\Delta x} - \frac{B_x(i,j,k)-B_x(i,j-1,k)}{\Delta y}\right]. \tag{2.5c}$$

2.3 Second Magnetic Half-Step

$$H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k+1} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta z}\right] \tag{2.6a}$$

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{2}\left[\frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z} - \frac{E^{n+1}_{z,i+1,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta x}\right] \tag{2.6b}$$

$$H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{2}\left[\frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j+1,k} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta y}\right]. \tag{2.6c}$$

In the AMReX Yee grid, with vacuum media, the second magnetic field updates are implemented as

$$B_x(i,j,k) = B_x(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_z(i,j+1,k)-E_z(i,j,k)}{\Delta y} - \frac{E_y(i,j,k+1)-E_y(i,j,k)}{\Delta z}\right] \tag{2.7a}$$

$$B_y(i,j,k) = B_y(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_x(i,j,k+1)-E_x(i,j,k)}{\Delta z} - \frac{E_z(i+1,j,k)-E_z(i,j,k)}{\Delta x}\right] \tag{2.7b}$$

$$B_z(i,j,k) = B_z(i,j,k) - \frac{1}{2\Delta t}\left[\frac{E_y(i+1,j,k)-E_y(i,j,k)}{\Delta x} - \frac{E_x(i,j+1,k)-E_x(i,j,k)}{\Delta y}\right]. \tag{2.7c}$$

This explicit method is simple and local, but the time step is restricted by the CFL stability condition. For a uniform lossless grid, a typical stability limit is

$$\Delta t \le \frac{1}{c\sqrt{\frac{1}{\Delta x^2}+\frac{1}{\Delta y^2}+\frac{1}{\Delta z^2}}}, \qquad c = \frac{1}{\sqrt{\mu\varepsilon}}. \tag{2.8}$$

3 ADI Method

The ADI-FDTD update advances the fields by a full time step Δt in two half steps of size $\Delta t/2$. Unlike the explicit three-stage leapfrog update above, each ADI half step first updates $\mathbf{E}$ implicitly over $\Delta t/2$, then updates $\mathbf{H}$ explicitly over $\Delta t/2$ from the updated electric field. The first half step treats one set of implicit directions; the second half step treats the complementary directions to complete the cycle. Define the half-step update coefficients

$$C_a = \frac{4\varepsilon - \sigma\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad C_b = \frac{2\Delta t}{4\varepsilon + \sigma\Delta t}, \qquad D_b = \frac{\Delta t}{2\mu}, \tag{3.1}$$

where Δt is the time step, ε is the permittivity, μ is the permeability, and σ is the conductivity. These differ from the full-step coefficients in (2.1) because each electric update uses $\Delta t/2$.

3.1 First ADI Half-Step

E_x and H_z By equation (1.2a),

$$\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k}\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + E^{n}_{x,i+\frac{1}{2},j,k}}{2} = [H_1], \tag{3.2}$$

where

$$[H_1] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.3}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_x^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_x^{n} + [H_1]. \tag{3.4}$$

Hence

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\, [H_1], \tag{3.5}$$

By equation (1.3c),

$$\begin{aligned} H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k} \Bigg(&\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ &- \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \end{aligned} \tag{3.6}$$

Substituting the magnetic update into (3.5) and eliminating $H_z^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\Bigg[H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta x}\right) \\ &\qquad - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k} - D_{b,i+\frac{1}{2},j-\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}}{\Delta y} - \frac{E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}}{\Delta x}\right)\Bigg] \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.7}$$

Collecting the unknown $E_x^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} \\ &+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2} + \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\right] E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}\, E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} \\ = &\; C_{a,i+\frac{1}{2},j,k}\, E^{n}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y\, \Delta x}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}\, D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y\, \Delta x}\left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.8}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned}
-\alpha_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k} + \beta_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} - \gamma_{x1} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} = {} & p_{x1} E^{n}_{x,i+\frac{1}{2},j,k} \\
& + \frac{1}{\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\
& - \frac{1}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\
& + \frac{q_{x1}}{\Delta x \Delta y}\left(E^{n}_{y,i+1,j-\frac{1}{2},k} - E^{n}_{y,i,j-\frac{1}{2},k}\right) \\
& - \frac{r_{x1}}{\Delta x \Delta y}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right),
\end{aligned} \tag{3.9}$$

and the magnetic update can be written as

$$H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + \frac{r_{x1}}{\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right) - \frac{r_{x1}}{\Delta x}\left(E^{n}_{y,i+1,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}\right). \tag{3.10}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{3.11}$$

$$\beta_{x1} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{3.12}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{3.13}$$

$$p_{x1} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.14}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{3.15}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.16}$$

E_y and H_x By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k} \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + E^{n}_{y,i,j+\frac{1}{2},k}}{2} = [H_2], \tag{3.17}$$

where

$$[H_2] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.18}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_y^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_y^{n} + [H_2]. \tag{3.19}$$

Hence

$$E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k} E^{n}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k} [H_2], \tag{3.20}$$

By equation (1.3a),

$$\begin{aligned}
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Bigg(& \frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\
& - \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y}\Bigg),
\end{aligned} \tag{3.21}$$

Substituting the magnetic update into (3.20) and eliminating $H_x^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} &= C_{a,i,j+\frac{1}{2},k}\, E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\Bigg[H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \right) \\
&\qquad - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} - D_{b,i,j+\frac{1}{2},k-\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1}}{\Delta z} - \frac{E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}}}{\Delta y} \right) \Bigg] \\
&\quad - \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right).
\end{aligned} \tag{3.22}
$$

Collecting the unknown $E_y^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} \\
&+ \left[1 + \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} \right] E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} \\
={}& C_{a,i,j+\frac{1}{2},k}\, E^{n}_{y,i,j+\frac{1}{2},k} + \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z} \left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z\, \Delta y} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right) \\
&+ \frac{C_{b,i,j+\frac{1}{2},k}\, D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z\, \Delta y} \left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}} \right) \\
&- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right).
\end{aligned} \tag{3.23}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k-1} + \beta_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} - \gamma_{y1} E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} &= p_{y1} E^{n}_{y,i,j+\frac{1}{2},k} \\
&\quad + \frac{1}{\Delta z} \left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j+\frac{1}{2},k-\frac{1}{2}} \right) \\
&\quad - \frac{1}{\Delta x} \left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n}_{z,i-\frac{1}{2},j+\frac{1}{2},k} \right) \\
&\quad + \frac{q_{y1}}{\Delta y \Delta z} \left(E^{n}_{z,i,j+1,k-\frac{1}{2}} - E^{n}_{z,i,j,k-\frac{1}{2}} \right) \\
&\quad - \frac{r_{y1}}{\Delta y \Delta z} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right),
\end{aligned} \tag{3.24}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{y1}}{\Delta z} \left(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \right) - \frac{r_{y1}}{\Delta y} \left(E^{n}_{z,i,j+1,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}} \right). \tag{3.25}
$$

where

$$\alpha_{y1} = \frac{D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z^2}, \tag{3.26}$$

$$\beta_{y1} = \frac{1}{C_{b,i,j+\frac{1}{2},k}} + \alpha_{y1} + \gamma_{y1}, \tag{3.27}$$

$$\gamma_{y1} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta z^2}, \tag{3.28}$$

$$p_{y1} = \frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.29}$$

$$q_{y1} = D_{b,i,j+\frac{1}{2},k-\frac{1}{2}}, \tag{3.30}$$

$$r_{y1} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.31}$$

E_z **and** H_y By equation (1.2c),

$$\varepsilon_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^{n}_{z,i,j,k+\frac{1}{2}}}{\Delta t/2} + \sigma_{i,j,k+\frac{1}{2}} \frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + E^{n}_{z,i,j,k+\frac{1}{2}}}{2} = [H_3], \tag{3.32}$$

where

$$[H_3] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.33}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n} + [H_3]. \tag{3.34}$$

Hence

$$E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^{n}_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_3], \tag{3.35}$$

By equation (1.3b),

$$\begin{aligned} H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \Bigg(& \frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} \\ & - \frac{E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta z} \Bigg), \end{aligned} \tag{3.36}$$

Substituting the magnetic update into (3.35) and eliminating $H_y^{n+\frac{1}{2}}$ terms gives

$$\begin{aligned} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} &= C_{a,i,j,k+\frac{1}{2}}\, E^{n}_{z,i,j,k+\frac{1}{2}} \\ &\quad + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Bigg[H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n}_{x,i+\frac{1}{2},j,k+1} - E^{n}_{x,i+\frac{1}{2},j,k}}{\Delta z} \right) \\ &\qquad - H^{n}_{y,i-\frac{1}{2},j,k+\frac{1}{2}} - D_{b,i-\frac{1}{2},j,k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n}_{x,i-\frac{1}{2},j,k+1} - E^{n}_{x,i-\frac{1}{2},j,k}}{\Delta z} \right) \Bigg] \\ &\quad - \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \left(H^{n}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \right). \end{aligned} \tag{3.37}$$

Collecting the unknown $E_z^{n+\frac{1}{2}}$ terms gives

$$
\begin{aligned}
&- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\, E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} \\
&+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2} + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\right] E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\
&- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}\, E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} \\
= {}& C_{a,i,j,k+\frac{1}{2}}\, E^n_{z,i,j,k+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
&- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\, \Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}\right) \\
&+ \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x\, \Delta z}\left(E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k}\right) \\
&- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right).
\end{aligned}
\tag{3.38}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z1} E^{n+\frac{1}{2}}_{z,i-1,j,k+\frac{1}{2}} + \beta_{z1} E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} - \gamma_{z1} E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} = {}& p_{z1} E^n_{z,i,j,k+\frac{1}{2}} \\
&+ \frac{1}{\Delta x}\left(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^n_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\right) \\
&- \frac{1}{\Delta y}\left(H^n_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^n_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\right) \\
&+ \frac{q_{z1}}{\Delta z \Delta x}\left(E^n_{x,i-\frac{1}{2},j,k+1} - E^n_{x,i-\frac{1}{2},j,k}\right) \\
&- \frac{r_{z1}}{\Delta z \Delta x}\left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}\right),
\end{aligned}
\tag{3.39}
$$

and the magnetic update can be written as

$$
H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{z1}}{\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) - \frac{r_{z1}}{\Delta z}\left(E^n_{x,i+\frac{1}{2},j,k+1} - E^n_{x,i+\frac{1}{2},j,k}\right). \tag{3.40}
$$

where

$$\alpha_{z1} = \frac{D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.41}$$

$$\beta_{z1} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z1} + \gamma_{z1}, \tag{3.42}$$

$$\gamma_{z1} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x^2}, \tag{3.43}$$

$$p_{z1} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.44}$$

$$q_{z1} = D_{b,i-\frac{1}{2},j,k+\frac{1}{2}}, \tag{3.45}$$

$$r_{z1} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.46}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$
\begin{aligned}
-E_x^{n+\frac{1}{2}}(i,j-1,k) + \left(2+\frac{4\Delta y^2}{c^2\Delta t^2}\right)E_x^{n+\frac{1}{2}}(i,j,k) - E_x^{n+\frac{1}{2}}(i,j+1,k) &= \frac{4\Delta y^2}{c^2\Delta t^2}E_x^n(i,j,k) \\
&+ \frac{2\Delta y^2}{\Delta t}\left[\frac{B_z^n(i,j,k)-B_z^n(i,j-1,k)}{\Delta y} - \frac{B_y^n(i,j,k)-B_y^n(i,j,k-1)}{\Delta z}\right] \\
&+ \Delta y\left[\frac{E_y^n(i+1,j-1,k)-E_y^n(i,j-1,k)}{\Delta x} - \frac{E_y^n(i+1,j,k)-E_y^n(i,j,k)}{\Delta x}\right],
\end{aligned}
\tag{3.47a}
$$

$$
\begin{aligned}
-E_y^{n+\frac{1}{2}}(i,j,k-1) + \left(2+\frac{4\Delta z^2}{c^2\Delta t^2}\right)E_y^{n+\frac{1}{2}}(i,j,k) - E_y^{n+\frac{1}{2}}(i,j,k+1) &= \frac{4\Delta z^2}{c^2\Delta t^2}E_y^n(i,j,k) \\
&+ \frac{2\Delta z^2}{\Delta t}\left[\frac{B_x^n(i,j,k)-B_x^n(i,j,k-1)}{\Delta z} - \frac{B_z^n(i,j,k)-B_z^n(i-1,j,k)}{\Delta x}\right] \\
&+ \Delta z\left[\frac{E_z^n(i,j+1,k-1)-E_z^n(i,j,k-1)}{\Delta y} - \frac{E_z^n(i,j+1,k)-E_z^n(i,j,k)}{\Delta y}\right],
\end{aligned}
\tag{3.47b}
$$

$$
\begin{aligned}
-E_z^{n+\frac{1}{2}}(i-1,j,k) + \left(2+\frac{4\Delta x^2}{c^2\Delta t^2}\right)E_z^{n+\frac{1}{2}}(i,j,k) - E_z^{n+\frac{1}{2}}(i+1,j,k) &= \frac{4\Delta x^2}{c^2\Delta t^2}E_z^n(i,j,k) \\
&+ \frac{2\Delta x^2}{\Delta t}\left[\frac{B_y^n(i,j,k)-B_y^n(i-1,j,k)}{\Delta x} - \frac{B_x^n(i,j,k)-B_x^n(i,j-1,k)}{\Delta y}\right] \\
&+ \Delta x\left[\frac{E_x^n(i-1,j,k+1)-E_x^n(i-1,j,k)}{\Delta z} - \frac{E_x^n(i,j,k+1)-E_x^n(i,j,k)}{\Delta z}\right],
\end{aligned}
\tag{3.47c}
$$

$$
B_x^{n+\frac{1}{2}}(i,j,k) = B_x^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1)-E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^n(i,j+1,k)-E_z^n(i,j,k)}{\Delta y}\right], \tag{3.47d}
$$

$$
B_y^{n+\frac{1}{2}}(i,j,k) = B_y^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k)-E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^n(i,j,k+1)-E_x^n(i,j,k)}{\Delta z}\right], \tag{3.47e}
$$

$$
B_z^{n+\frac{1}{2}}(i,j,k) = B_z^n(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k)-E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^n(i+1,j,k)-E_y^n(i,j,k)}{\Delta x}\right]. \tag{3.47f}
$$

3.2 Second ADI Half-Step

E_x and H_y By equation (1.2a),

$$
\varepsilon_{i+\frac{1}{2},j,k}\frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta t/2} + \sigma_{i+\frac{1}{2},j,k}\frac{E^{n+1}_{x,i+\frac{1}{2},j,k} + E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{2} = [H_4], \tag{3.48}
$$

where

$$
[H_4] = \frac{H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y} - \frac{H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+1}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z}. \tag{3.49}
$$

Rearranging gives

$$
\left(\frac{2\varepsilon}{\Delta t}+\frac{\sigma}{2}\right)E_x^{n+1} = \left(\frac{2\varepsilon}{\Delta t}-\frac{\sigma}{2}\right)E_x^{n+\frac{1}{2}} + [H_4]. \tag{3.50}
$$

Hence

$$
E^{n+1}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\,E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_4], \tag{3.51}
$$

By equation (1.3b),

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}\Bigg(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z}\Bigg), \tag{3.52}$$

Substituting the magnetic update into (3.51) and eliminating H_y^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{x,i+\frac{1}{2},j,k} &= C_{a,i+\frac{1}{2},j,k} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &+ \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\Bigg[H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}}{\Delta z}\right) \\ &\qquad - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}} - D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}\left(\frac{E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}}{\Delta x} - \frac{E^{n+1}_{x,i+\frac{1}{2},j,k} - E^{n+1}_{x,i+\frac{1}{2},j,k-1}}{\Delta z}\right)\Bigg]. \end{aligned} \tag{3.53}$$

Collecting the unknown E_x^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} E^{n+1}_{x,i+\frac{1}{2},j,k+1} \\ &+ \left[1 + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2} + \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}\right] E^{n+1}_{x,i+\frac{1}{2},j,k} \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2} E^{n+1}_{x,i+\frac{1}{2},j,k-1} \\ &= C_{a,i+\frac{1}{2},j,k} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\ &- \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z\,\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) \\ &+ \frac{C_{b,i+\frac{1}{2},j,k} D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z\,\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}\right). \end{aligned} \tag{3.54}$$

Equivalently, the tridiagonal system can be written as

$$\begin{aligned} -\alpha_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k-1} + \beta_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k} - \gamma_{x2} E^{n+1}_{x,i+\frac{1}{2},j,k+1} &= p_{x2} E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} \\ &+ \frac{1}{\Delta y}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right) \\ &- \frac{1}{\Delta z}\left(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right) \\ &+ \frac{q_{x2}}{\Delta x \Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k-\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k-\frac{1}{2}}\right) \\ &- \frac{r_{x2}}{\Delta x \Delta z}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right), \end{aligned} \tag{3.55}$$

and the magnetic update can be written as

$$H^{n+1}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} + \frac{r_{x2}}{\Delta x}\left(E^{n+\frac{1}{2}}_{z,i+1,j,k+\frac{1}{2}} - E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}\right) - \frac{r_{x2}}{\Delta z}\left(E^{n+1}_{x,i+\frac{1}{2},j,k+1} - E^{n+1}_{x,i+\frac{1}{2},j,k}\right). \tag{3.56}$$

where

$$\alpha_{x2} = \frac{D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}}{\Delta z^2}, \tag{3.57}$$

$$\beta_{x2} = \frac{1}{C_{b,i+\frac{1}{2},j,k}} + \alpha_{x2} + \gamma_{x2}, \tag{3.58}$$

$$\gamma_{x2} = \frac{D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}}{\Delta z^2}, \tag{3.59}$$

$$p_{x2} = \frac{C_{a,i+\frac{1}{2},j,k}}{C_{b,i+\frac{1}{2},j,k}}, \tag{3.60}$$

$$q_{x2} = D_{b,i+\frac{1}{2},j,k-\frac{1}{2}}, \tag{3.61}$$

$$r_{x2} = D_{b,i+\frac{1}{2},j,k+\frac{1}{2}}. \tag{3.62}$$

E_y and H_z By equation (1.2b),

$$\varepsilon_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta t/2} + \sigma_{i,j+\frac{1}{2},k}\frac{E^{n+1}_{y,i,j+\frac{1}{2},k} + E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{2} = [H_5], \tag{3.63}$$

where

$$[H_5] = \frac{H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}}{\Delta z} - \frac{H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} - H^{n+1}_{z,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x}. \tag{3.64}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E^{n+1}_y = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E^{n+\frac{1}{2}}_y + [H_5]. \tag{3.65}$$

Hence

$$E^{n+1}_{y,i,j+\frac{1}{2},k} = C_{a,i,j+\frac{1}{2},k}\, E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} + C_{b,i,j+\frac{1}{2},k}\, [H_5], \tag{3.66}$$

By equation (1.3c),

$$\begin{aligned} H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k} = H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\Bigg(&\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} \\ &- \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\Bigg), \end{aligned} \tag{3.67}$$

Substituting the magnetic update into (3.66) and eliminating H^{n+1}_z terms gives

$$\begin{aligned} E^{n+1}_{y,i,j+\frac{1}{2},k} = {}& C_{a,i,j+\frac{1}{2},k}\, E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \\ &+ \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right) \\ &- \frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\Bigg[H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k} + D_{b,i+\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i+1,j+\frac{1}{2},k} - E^{n+1}_{y,i,j+\frac{1}{2},k}}{\Delta x}\right) \\ &\qquad - H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k} - D_{b,i-\frac{1}{2},j+\frac{1}{2},k}\left(\frac{E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k} - E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}}{\Delta y} - \frac{E^{n+1}_{y,i,j+\frac{1}{2},k} - E^{n+1}_{y,i-1,j+\frac{1}{2},k}}{\Delta x}\right)\Bigg]. \end{aligned} \tag{3.68}$$

Collecting the unknown E_y^{n+1} terms gives

$$
\begin{aligned}
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\,E^{n+1}_{y,i+1,j+\frac{1}{2},k}\\
&+\left[1+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\right]E^{n+1}_{y,i,j+\frac{1}{2},k}\\
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}\,E^{n+1}_{y,i-1,j+\frac{1}{2},k}\\
={}&C_{a,i,j+\frac{1}{2},k}\,E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}+\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&-\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)\\
&+\frac{C_{b,i,j+\frac{1}{2},k}\,D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x\,\Delta y}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right).
\end{aligned}
\tag{3.69}
$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{y2}E^{n+1}_{y,i-1,j+\frac{1}{2},k}+\beta_{y2}E^{n+1}_{y,i,j+\frac{1}{2},k}-\gamma_{y2}E^{n+1}_{y,i+1,j+\frac{1}{2},k}={}&p_{y2}E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\\
&+\frac{1}{\Delta z}\left(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}}-H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k-\frac{1}{2}}\right)\\
&-\frac{1}{\Delta x}\left(H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n+\frac{1}{2}}_{z,i-\frac{1}{2},j+\frac{1}{2},k}\right)\\
&+\frac{q_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i-\frac{1}{2},j,k}\right)\\
&-\frac{r_{y2}}{\Delta y\Delta x}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right),
\end{aligned}
\tag{3.70}
$$

and the magnetic update can be written as

$$
H^{n+1}_{z,i+\frac{1}{2},j+\frac{1}{2},k}=H^{n+\frac{1}{2}}_{z,i+\frac{1}{2},j+\frac{1}{2},k}+\frac{r_{y2}}{\Delta y}\left(E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}-E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}\right)-\frac{r_{y2}}{\Delta x}\left(E^{n+1}_{y,i+1,j+\frac{1}{2},k}-E^{n+1}_{y,i,j+\frac{1}{2},k}\right).
\tag{3.71}
$$

where

$$\alpha_{y2}=\frac{D_{b,i-\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.72}$$

$$\beta_{y2}=\frac{1}{C_{b,i,j+\frac{1}{2},k}}+\alpha_{y2}+\gamma_{y2}, \tag{3.73}$$

$$\gamma_{y2}=\frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta x^2}, \tag{3.74}$$

$$p_{y2}=\frac{C_{a,i,j+\frac{1}{2},k}}{C_{b,i,j+\frac{1}{2},k}}, \tag{3.75}$$

$$q_{y2}=D_{b,i-\frac{1}{2},j+\frac{1}{2},k}, \tag{3.76}$$

$$r_{y2}=D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{3.77}$$

E_z **and** H_x By equation (1.2c),

$$
\varepsilon_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}-E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{\Delta t/2}+\sigma_{i,j,k+\frac{1}{2}}\frac{E^{n+1}_{z,i,j,k+\frac{1}{2}}+E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}}}{2}=[H_6],
\tag{3.78}
$$

where

$$[H_6] = \frac{H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x} - \frac{H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+1}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y}. \tag{3.79}$$

Rearranging gives

$$\left(\frac{2\varepsilon}{\Delta t} + \frac{\sigma}{2}\right) E_z^{n+1} = \left(\frac{2\varepsilon}{\Delta t} - \frac{\sigma}{2}\right) E_z^{n+\frac{1}{2}} + [H_6]. \tag{3.80}$$

Hence

$$E^{n+1}_{z,i,j,k+\frac{1}{2}} = C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + C_{b,i,j,k+\frac{1}{2}}\, [H_6], \tag{3.81}$$

By equation (1.3a),

$$\begin{aligned} H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \Big(&\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} \\ &- \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \Big), \end{aligned} \tag{3.82}$$

Substituting the magnetic update into (3.81) and eliminating H_x^{n+1} terms gives

$$\begin{aligned} E^{n+1}_{z,i,j,k+\frac{1}{2}} = {}& C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Big(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \Big) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \Bigg[H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + D_{b,i,j+\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}}{\Delta y} \right) \\ &\qquad - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} - D_{b,i,j-\frac{1}{2},k+\frac{1}{2}} \left(\frac{E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}}{\Delta z} - \frac{E^{n+1}_{z,i,j,k+\frac{1}{2}} - E^{n+1}_{z,i,j-1,k+\frac{1}{2}}}{\Delta y} \right) \Bigg]. \end{aligned} \tag{3.83}$$

Collecting the unknown E_z^{n+1} terms gives

$$\begin{aligned} &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j+1,k+\frac{1}{2}} \\ &+ \left[1 + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} + \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2} \right] E^{n+1}_{z,i,j,k+\frac{1}{2}} \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}\, E^{n+1}_{z,i,j-1,k+\frac{1}{2}} \\ = {}& C_{a,i,j,k+\frac{1}{2}}\, E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} + \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta x} \Big(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}} \Big) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}}{\Delta y} \Big(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}} \Big) \\ &- \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z} \Big(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k} \Big) \\ &+ \frac{C_{b,i,j,k+\frac{1}{2}}\, D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y\, \Delta z} \Big(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k} \Big). \end{aligned} \tag{3.84}$$

Equivalently, the tridiagonal system can be written as

$$
\begin{aligned}
-\alpha_{z2}E^{n+1}_{z,i,j-1,k+\frac{1}{2}} + \beta_{z2}E^{n+1}_{z,i,j,k+\frac{1}{2}} - \gamma_{z2}E^{n+1}_{z,i,j+1,k+\frac{1}{2}} = p_{z2}E^{n+\frac{1}{2}}_{z,i,j,k+\frac{1}{2}} &\\
+ \frac{1}{\Delta x}\Big(H^{n+\frac{1}{2}}_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{y,i-\frac{1}{2},j,k+\frac{1}{2}}\Big) &\\
- \frac{1}{\Delta y}\Big(H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} - H^{n+\frac{1}{2}}_{x,i,j-\frac{1}{2},k+\frac{1}{2}}\Big) &\\
+ \frac{q_{z2}}{\Delta z\Delta y}\Big(E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j-\frac{1}{2},k}\Big) &\\
- \frac{r_{z2}}{\Delta z\Delta y}\Big(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\Big), &
\end{aligned}
\tag{3.85}
$$

and the magnetic update can be written as

$$
H^{n+1}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} = H^{n+\frac{1}{2}}_{x,i,j+\frac{1}{2},k+\frac{1}{2}} + \frac{r_{z2}}{\Delta z}\Big(E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k+1} - E^{n+\frac{1}{2}}_{y,i,j+\frac{1}{2},k}\Big) - \frac{r_{z2}}{\Delta y}\Big(E^{n+1}_{z,i,j+1,k+\frac{1}{2}} - E^{n+1}_{z,i,j,k+\frac{1}{2}}\Big). \tag{3.86}
$$

where

$$\alpha_{z2} = \frac{D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.87}$$

$$\beta_{z2} = \frac{1}{C_{b,i,j,k+\frac{1}{2}}} + \alpha_{z2} + \gamma_{z2}, \tag{3.88}$$

$$\gamma_{z2} = \frac{D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}}{\Delta y^2}, \tag{3.89}$$

$$p_{z2} = \frac{C_{a,i,j,k+\frac{1}{2}}}{C_{b,i,j,k+\frac{1}{2}}}, \tag{3.90}$$

$$q_{z2} = D_{b,i,j-\frac{1}{2},k+\frac{1}{2}}, \tag{3.91}$$

$$r_{z2} = D_{b,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{3.92}$$

In the AMReX Yee grid, with vacuum media, the field updates are implemented as

$$
\begin{aligned}
-E_x^{n+1}(i,j,k-1) + \left(2 + \frac{4\Delta z^2}{c^2\Delta t^2}\right)E_x^{n+1}(i,j,k) - E_x^{n+1}(i,j,k+1) = \frac{4\Delta z^2}{c^2\Delta t^2}E_x^{n+\frac{1}{2}}(i,j,k) &\\
+ \frac{2\Delta z^2}{\Delta t}\left[\frac{B_z^{n+\frac{1}{2}}(i,j,k) - B_z^{n+\frac{1}{2}}(i,j-1,k)}{\Delta y} - \frac{B_y^{n+\frac{1}{2}}(i,j,k) - B_y^{n+\frac{1}{2}}(i,j,k-1)}{\Delta z}\right] &\\
+ \Delta z\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k-1) - E_z^{n+\frac{1}{2}}(i,j,k-1)}{\Delta x} - \frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x}\right], &
\end{aligned}
\tag{3.93a}
$$

$$
\begin{aligned}
-E_y^{n+1}(i-1,j,k) + \left(2 + \frac{4\Delta x^2}{c^2\Delta t^2}\right)E_y^{n+1}(i,j,k) - E_y^{n+1}(i+1,j,k) = \frac{4\Delta x^2}{c^2\Delta t^2}E_y^{n+\frac{1}{2}}(i,j,k) &\\
+ \frac{2\Delta x^2}{\Delta t}\left[\frac{B_x^{n+\frac{1}{2}}(i,j,k) - B_x^{n+\frac{1}{2}}(i,j,k-1)}{\Delta z} - \frac{B_z^{n+\frac{1}{2}}(i,j,k) - B_z^{n+\frac{1}{2}}(i-1,j,k)}{\Delta x}\right] &\\
+ \Delta x\left[\frac{E_x^{n+\frac{1}{2}}(i-1,j+1,k) - E_x^{n+\frac{1}{2}}(i-1,j,k)}{\Delta y} - \frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y}\right], &
\end{aligned}
\tag{3.93b}
$$

$$
\begin{aligned}
-E_z^{n+1}(i,j-1,k) + \left(2 + \frac{4\Delta y^2}{c^2\Delta t^2}\right) E_z^{n+1}(i,j,k) - E_z^{n+1}(i,j+1,k) = \frac{4\Delta y^2}{c^2\Delta t^2} E_z^{n+\frac{1}{2}}(i,j,k) \\
+ \frac{2\Delta y^2}{\Delta t}\left[\frac{B_y^{n+\frac{1}{2}}(i,j,k) - B_y^{n+\frac{1}{2}}(i-1,j,k)}{\Delta x} - \frac{B_x^{n+\frac{1}{2}}(i,j,k) - B_x^{n+\frac{1}{2}}(i,j-1,k)}{\Delta y}\right] \\
+ \Delta y\left[\frac{E_y^{n+\frac{1}{2}}(i,j-1,k+1) - E_y^{n+\frac{1}{2}}(i,j-1,k)}{\Delta z} - \frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z}\right],
\end{aligned}
\tag{3.93c}
$$

$$
B_x^{n+1}(i,j,k) = B_x^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_y^{n+\frac{1}{2}}(i,j,k+1) - E_y^{n+\frac{1}{2}}(i,j,k)}{\Delta z} - \frac{E_z^{n+1}(i,j+1,k) - E_z^{n+1}(i,j,k)}{\Delta y}\right], \tag{3.93d}
$$

$$
B_y^{n+1}(i,j,k) = B_y^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_z^{n+\frac{1}{2}}(i+1,j,k) - E_z^{n+\frac{1}{2}}(i,j,k)}{\Delta x} - \frac{E_x^{n+1}(i,j,k+1) - E_x^{n+1}(i,j,k)}{\Delta z}\right], \tag{3.93e}
$$

$$
B_z^{n+1}(i,j,k) = B_z^{n+\frac{1}{2}}(i,j,k) + \frac{\Delta t}{2}\left[\frac{E_x^{n+\frac{1}{2}}(i,j+1,k) - E_x^{n+\frac{1}{2}}(i,j,k)}{\Delta y} - \frac{E_y^{n+1}(i+1,j,k) - E_y^{n+1}(i,j,k)}{\Delta x}\right]. \tag{3.93f}
$$

4 Solution of Linear Systems

See Tridiagonal matrix algorithm for more details.

5 Perfect Electric Conductor Boundary Conditions

For a perfect electric conductor (PEC) boundary with outward unit normal $\hat{\mathbf{n}}$, the electric field satisfies

$$
\hat{\mathbf{n}} \times \mathbf{E} = 0. \tag{5.1}
$$

That is, the tangential electric field vanishes on the conducting surface. If the field inside the conductor is not evolved and the initial data are consistent with the boundary, then Faraday's law also implies that the normal magnetic flux remains zero on the wall,

$$
\hat{\mathbf{n}} \cdot \mathbf{B} = 0. \tag{5.2}
$$

Now consider a rectangular domain with PEC boundaries on the two faces normal to x, namely the planes $x = x_{\mathrm{lo}}$ and $x = x_{\mathrm{hi}}$, while the four faces normal to y and z remain periodic. On the AMReX Yee grid, E_y and E_z are nodal in x and therefore live directly on the x-boundary planes, whereas E_x is centered at $i + \frac{1}{2}$ in the x direction and is not stored on the boundary itself. If the domain has N_x cells in x, the discrete PEC conditions are therefore

$$
E_y(0,j,k) = E_y(N_x,j,k) = 0, \qquad E_z(0,j,k) = E_z(N_x,j,k) = 0. \tag{5.3}
$$

No value is imposed directly on E_x at the wall, since it is the normal component.
In the Yee update, the implementation is to overwrite the tangential electric fields on the two PEC planes after each electric-field stage,

$$
E_y^m(0,j,k) = E_y^m(N_x,j,k) = 0, \qquad E_z^m(0,j,k) = E_z^m(N_x,j,k) = 0, \tag{5.4}
$$

where $m = n + \frac{1}{2}$ or $n+1$ depending on the stage. The neighboring magnetic updates then use differences against these zero boundary values. For example, the x-derivative appearing in the B_y update is approximated by

$$
\left.\frac{\partial E_z}{\partial x}\right|_{i+\frac{1}{2},j,k} \approx \frac{E_z(i+1,j,k) - E_z(i,j,k)}{\Delta x}. \tag{5.5}
$$

At the first interior face next to $x = x_{\mathrm{lo}}$,

$$
\left.\frac{\partial E_z}{\partial x}\right|_{\frac{1}{2},j,k} \approx \frac{E_z(1,j,k) - E_z(0,j,k)}{\Delta x} = \frac{E_z(1,j,k)}{\Delta x}, \tag{5.6}
$$

while at the last interior face next to $x = x_{\mathrm{hi}}$,

$$
\left.\frac{\partial E_z}{\partial x}\right|_{N_x-\frac{1}{2},j,k} \approx \frac{E_z(N_x,j,k) - E_z(N_x-1,j,k)}{\Delta x} = -\frac{E_z(N_x-1,j,k)}{\Delta x}. \tag{5.7}
$$

Likewise, the update for the normal magnetic component B_x on the PEC planes involves only tangential derivatives of E_y and E_z, so if B_x is initialized consistently on the wall then it remains zero there.
In the ADI scheme, the same PEC condition enters as a Dirichlet condition whenever a tangential electric component is solved implicitly along x. Because the y and z directions remain periodic, the implicit solves along those directions are unchanged and stay cyclic. The only modified line solves are the x-directed ones:

- the E_z solve in the first half-step, equation (3.39);
- the E_y solve in the second half-step, equation (3.70).

For the first half-step, the E_z solve uses

$$E^{n+\frac{1}{2}}_{z,0,j,k+\frac{1}{2}} = 0, \qquad E^{n+\frac{1}{2}}_{z,N_x,j,k+\frac{1}{2}} = 0. \tag{5.8}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{z1} E^{n+\frac{1}{2}}_{z,1,j,k+\frac{1}{2}} - \gamma_{z1} E^{n+\frac{1}{2}}_{z,2,j,k+\frac{1}{2}} = \text{RHS}, \tag{5.9}$$

and

$$-\alpha_{z1} E^{n+\frac{1}{2}}_{z,N_x-2,j,k+\frac{1}{2}} + \beta_{z1} E^{n+\frac{1}{2}}_{z,N_x-1,j,k+\frac{1}{2}} = \text{RHS}, \tag{5.10}$$

respectively. The boundary unknowns do not participate in the solve; they are prescribed by the PEC condition.
Similarly, in the second half-step the E_y solve uses

$$E^{n+1}_{y,0,j+\frac{1}{2},k} = 0, \qquad E^{n+1}_{y,N_x,j+\frac{1}{2},k} = 0. \tag{5.11}$$

Hence the first and last interior tridiagonal equations are

$$\beta_{y2} E^{n+1}_{y,1,j+\frac{1}{2},k} - \gamma_{y2} E^{n+1}_{y,2,j+\frac{1}{2},k} = \text{RHS}, \tag{5.12}$$

and

$$-\alpha_{y2} E^{n+1}_{y,N_x-2,j+\frac{1}{2},k} + \beta_{y2} E^{n+1}_{y,N_x-1,j+\frac{1}{2},k} = \text{RHS}, \tag{5.13}$$

respectively.
Note that we do not need to explicitly impose $\hat{\mathbf{n}} \cdot \mathbf{B} = 0$ separately. For an x-normal wall, the magnetic update on the boundary plane $i = i_b$, with $i_b \in \{0, N_x\}$, is

$$B_x^{n+1}(i_b, j, k) = B_x^n(i_b, j, k) + \frac{\Delta t}{2} \left[\frac{E_y^{n+\frac{1}{2}}(i_b, j, k+1) - E_y^{n+\frac{1}{2}}(i_b, j, k)}{\Delta z} - \frac{E_z^{n+1}(i_b, j+1, k) - E_z^{n+1}(i_b, j, k)}{\Delta y} \right]. \tag{5.14}$$

Since the PEC condition gives

$$E_y(i_b, j, k) = 0, \qquad E_z(i_b, j, k) = 0 \tag{5.15}$$

for all j and k, it follows that

$$B_x^{n+1}(i_b, j, k) = B_x^n(i_b, j, k). \tag{5.16}$$

Therefore, if

$$B_x^0(i_b, j, k) = 0, \tag{5.17}$$

then

$$B_x^n(i_b, j, k) = 0 \tag{5.18}$$

is already satisfied for all n.

6 Dispersion Analysis

We analyze the homogeneous-vacuum ADI scheme for a wave propagating in the x-direction with

$$E_x = E_y = 0, \qquad H_x = H_z = 0, \tag{6.1}$$

so that the only nontrivial fields are E_z and H_y. The other axis-aligned polarizations follow by a cyclic permutation of coordinates.

For simplicity, we consider the case of vacuum media. Set $\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$, and

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}. \tag{6.2}$$

The half-step coefficients (3.1) are then

$$C_a = 1, \qquad C_b = \frac{\Delta t}{2\varepsilon_0}, \qquad D_b = \frac{\Delta t}{2\mu_0}. \tag{6.3}$$

Define the coefficient γ by

$$\gamma := C_b D_b = \frac{(c\Delta t)^2}{4}. \tag{6.4}$$

Define the two staggered first-difference operators by

$$\left(\delta_x^E E_z\right)_{i+\frac{1}{2},j,k+\frac{1}{2}} := \frac{E_{z,i+1,j,k+\frac{1}{2}} - E_{z,i,j,k+\frac{1}{2}}}{\Delta x}, \tag{6.5}$$

$$\left(\delta_x^H H_y\right)_{i,j,k+\frac{1}{2}} := \frac{H_{y,i+\frac{1}{2},j,k+\frac{1}{2}} - H_{y,i-\frac{1}{2},j,k+\frac{1}{2}}}{\Delta x}. \tag{6.6}$$

Thus δ_x^E maps the E_z grid to the H_y grid, while δ_x^H maps the H_y grid to the E_z grid. Their composition is the centered second difference

$$\left(\delta_x^2 E_z\right)_{i,j,k+\frac{1}{2}} := \left(\delta_x^H \delta_x^E E_z\right)_{i,j,k+\frac{1}{2}} = \frac{E_{z,i-1,j,k+\frac{1}{2}} - 2E_{z,i,j,k+\frac{1}{2}} + E_{z,i+1,j,k+\frac{1}{2}}}{\Delta x^2}. \tag{6.7}$$

6.1 One-dimensional ADI update

Restricting (3.39)–(3.40) to this polarization and multiplying the electric equation by C_b, the first half-step is

$$\left(1 - \gamma\delta_x^2\right) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n, \tag{6.8}$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}}, \tag{6.9}$$

where E_z is implicit along x. In the second half-step, E_z is formally implicit along y. Because this mode has no y-dependence, that second-difference term vanishes and the update reduces to

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}}, \tag{6.10}$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}}. \tag{6.11}$$

Both magnetic half-updates use the same intermediate electric field; therefore,

$$H_y^{n+1} = H_y^n + 2D_b \delta_x^E E_z^{n+\frac{1}{2}} \tag{6.12}$$

6.2 Plane-wave ansatz and Yee symbols

At any time level m, write a spatial Fourier mode on the staggered Yee grid as

$$E^m_{z,i,j,k+\frac{1}{2}} = \widehat{E}_z^m e^{\mathrm{i}k_x i\Delta x}, \tag{6.13}$$

$$H^m_{y,i+\frac{1}{2},j,k+\frac{1}{2}} = \widehat{H}_y^m e^{\mathrm{i}k_x (i+\frac{1}{2})\Delta x}. \tag{6.14}$$

Plugging into (6.5) gives

$$\left(\delta_x^E E_z^m\right)_{i+\frac{1}{2},j,k+\frac{1}{2}} = \widehat{E}_z^m e^{\mathrm{i}k_x(i+\frac{1}{2})\Delta x} \frac{e^{\mathrm{i}k_x\Delta x/2} - e^{-\mathrm{i}k_x\Delta x/2}}{\Delta x} \tag{6.15}$$

$$= \mathrm{i}K_x \widehat{E}_z^m e^{\mathrm{i}k_x(i+\frac{1}{2})\Delta x}, \tag{6.16}$$

where

$$K_x := \frac{2}{\Delta x} \sin\left(\frac{k_x \Delta x}{2}\right). \tag{6.17}$$

Similarly, (6.6) gives

$$\left(\delta_x^H H_y^m\right)_{i,j,k+\frac{1}{2}} = \widehat{H}_y^{\,m} e^{\mathrm{i}k_x i\Delta x}\frac{e^{\mathrm{i}k_x\Delta x/2}-e^{-\mathrm{i}k_x\Delta x/2}}{\Delta x} \tag{6.18}$$

$$= \mathrm{i}K_x\widehat{H}_y^{\,m} e^{\mathrm{i}k_x i\Delta x}. \tag{6.19}$$

Consequently, the Fourier symbols of the explicitly defined operators are

$$\delta_x^E \longleftrightarrow \mathrm{i}K_x, \qquad \delta_x^H \longleftrightarrow \mathrm{i}K_x, \qquad \delta_x^2 \longleftrightarrow -K_x^2. \tag{6.20}$$

As $\Delta x \to 0$, $K_x \to k_x$, so these symbols recover their continuum counterparts.

6.3 Fourier reduction of the half-steps

For brevity, define the derivative symbol and implicit denominator

$$s := \mathrm{i}K_x, \qquad D := 1+\gamma K_x^2. \tag{6.21}$$

Substituting into (6.8) and using (6.20), the first-half electric update becomes

$$D\widehat{E}_z^{\,n+\frac{1}{2}} = \widehat{E}_z^{\,n} + C_b s\widehat{H}_y^{\,n}. \tag{6.22}$$

Similarly for (6.9), we have

$$\widehat{H}_y^{\,n+\frac{1}{2}} = \widehat{H}_y^{\,n} + D_b s\widehat{E}_z^{\,n+\frac{1}{2}}. \tag{6.23}$$

Substituting into (6.10) and using (6.20), the second-half electric update becomes

$$\widehat{E}_z^{\,n+1} = \widehat{E}_z^{\,n+\frac{1}{2}} + C_b s\widehat{H}_y^{\,n+\frac{1}{2}}, \tag{6.24}$$

Similarly for (6.11), we have

$$\widehat{H}_y^{\,n+1} = \widehat{H}_y^{\,n} + 2D_b s\widehat{E}_z^{\,n+\frac{1}{2}} \tag{6.25}$$

6.4 Amplification matrix

Solving (6.22) for the intermediate electric amplitude gives

$$\widehat{E}_z^{\,n+\frac{1}{2}} = \frac{1}{D}\widehat{E}_z^{\,n} + \frac{C_b s}{D}\widehat{H}_y^{\,n}. \tag{6.26}$$

Substitution into (6.24) and (6.25), using $C_bD_b = \gamma$ and $s^2 = -K_x^2$, gives the one-step map

$$\begin{pmatrix}\widehat{E}_z\\ \widehat{H}_y\end{pmatrix}^{n+1} = G\begin{pmatrix}\widehat{E}_z\\ \widehat{H}_y\end{pmatrix}^{n}, \qquad G = \frac{1}{D}\begin{pmatrix}1-\gamma K_x^2 & 2C_b s\\ 2D_b s & 1-\gamma K_x^2\end{pmatrix}. \tag{6.27}$$

6.5 Numerical dispersion relation

A time-harmonic mode satisfies

$$\begin{pmatrix}\widehat{E}_z\\ \widehat{H}_y\end{pmatrix}^{n+1} = \lambda\begin{pmatrix}\widehat{E}_z\\ \widehat{H}_y\end{pmatrix}^{n}, \qquad \lambda = e^{-\mathrm{i}\omega\Delta t}. \tag{6.28}$$

The characteristic equation $\det(G-\lambda I) = 0$ yields

$$\lambda = \frac{1-\gamma K_x^2 \pm 2\mathrm{i}\,K_x\sqrt{\gamma}}{1+\gamma K_x^2}. \tag{6.29}$$

Introduce the CFL number $S := c\Delta t/\Delta x$ and

$$\nu := S\sin\left(\frac{k_x\Delta x}{2}\right) = K_x\sqrt{\gamma}. \tag{6.30}$$

Then $\gamma K_x^2 = \nu^2$, and the eigenvalues simplify to

$$\lambda = \frac{1\pm\mathrm{i}\nu}{1\mp\mathrm{i}\nu}. \tag{6.31}$$

The forward-propagating branch ($\omega > 0$) is

$$\lambda = \frac{1 - \mathrm{i}\nu}{1 + \mathrm{i}\nu}, \tag{6.32}$$

for which $|\lambda| = 1$ (unconditional stability of this one-dimensional reduction) and

$$\arg(\lambda) = -2\arctan(\nu). \tag{6.33}$$

Therefore the ADI numerical dispersion relation is

$$\boxed{\omega_{\mathrm{ADI}}(k_x) = \frac{2}{\Delta t}\,\arctan\!\left(S\sin\frac{k_x\Delta x}{2}\right).} \tag{6.34}$$

In the continuum limit $k_x\Delta x \to 0$ with fixed S, one has $\sin(k_x\Delta x/2) \approx k_x\Delta x/2$ and $\arctan\nu \approx \nu$, hence

$$\omega_{\mathrm{ADI}} \approx \frac{2}{\Delta t}\cdot S\cdot\frac{k_x\Delta x}{2} = ck_x, \tag{6.35}$$

as required. The leading truncation error is $O((k_x\Delta x)^2)$ at fixed Courant number, so the phase-velocity error decreases quadratically with cells-per-wavelength.

7 External Excitation

External field excitations are prescribed by user-defined grid functions of space and time, together with a spatial flag that selects the excitation type at each Yee edge (or face). Following the Artemis convention,

- flag = 0: no excitation,
- flag = 1: hard source (Dirichlet), field equals the prescribed value,
- flag = 2: soft source, the prescribed value is *added* to the field update.

In the macroscopic ADI push these two families are treated differently: soft electric sources are built into the implicit electric systems, while magnetic sources are applied after each magnetic half-step. Hard electric sources are not supported in the ADI electric solve.

7.1 Soft Electric Sources in the ADI Electric Update

Write $S^{n+\frac{1}{4}}_{E_x,i+\frac{1}{2},j,k}$ for the soft-source value assigned to the E_x edge at $\left(i+\frac{1}{2},j,k\right)$ and time level $n+\frac{1}{4}$ (and likewise for E_y and E_z at their staggered locations). Over a full step the intended soft contribution at that edge is split equally between the two ADI half-steps and evaluated at the quarter-step levels $n+\frac{1}{4}$ and $n+\frac{3}{4}$. The first-half electric update (3.5) is then modified to

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\,E^{n}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_1] + \frac{1}{2}\,S^{n+\frac{1}{4}}_{E_x,i+\frac{1}{2},j,k}, \tag{7.1}$$

on edges where the E_x flag equals 2, and is unchanged elsewhere. The second half-step (3.51) is analogous,

$$E^{n+1}_{x,i+\frac{1}{2},j,k} = C_{a,i+\frac{1}{2},j,k}\,E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k} + C_{b,i+\frac{1}{2},j,k}\,[H_4] + \frac{1}{2}\,S^{n+\frac{3}{4}}_{E_x,i+\frac{1}{2},j,k}. \tag{7.2}$$

The same construction applies componentwise to E_y and E_z. The magnetic updates themselves are unchanged by the electric source; only the electric right-hand side is affected.

7.2 Effect on the ADI Tridiagonal System

Dividing (7.1) by $C_{b,i+\frac{1}{2},j,k}$ and substituting the magnetic update as in Section 3 yields the same left-hand side as (3.9). On soft-source edges the only modification is an additional known term on the right-hand side:

$$\begin{aligned}-\alpha_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}+\beta_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}-\gamma_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k} &= p_{x1}E^{n}_{x,i+\frac{1}{2},j,k}\\ &+\frac{1}{\Delta y}\left(H^{n}_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^{n}_{z,i+\frac{1}{2},j-\frac{1}{2},k}\right)\\ &-\frac{1}{\Delta z}\left(H^{n}_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^{n}_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\right)\\ &+\frac{q_{x1}}{\Delta x\Delta y}\left(E^{n}_{y,i+1,j-\frac{1}{2},k}-E^{n}_{y,i,j-\frac{1}{2},k}\right)\\ &-\frac{r_{x1}}{\Delta x\Delta y}\left(E^{n}_{y,i+1,j+\frac{1}{2},k}-E^{n}_{y,i,j+\frac{1}{2},k}\right)\\ &+\frac{1}{2C_{b,i+\frac{1}{2},j,k}}S^{n+\frac{1}{4}}_{E_x,i+\frac{1}{2},j,k},\end{aligned} \tag{7.3}$$

with the same coefficients $\alpha_{x1},\beta_{x1},\gamma_{x1},p_{x1},q_{x1},r_{x1}$ as in Section 3. Away from soft-source edges the final term is omitted. In the implementation this contribution is the increment

$$\texttt{rhs}_{i,j,k} \mathrel{+}= \tfrac{1}{2}\,S^{n+\frac{1}{4}}_{E_x,i+\frac{1}{2},j,k}/C_{b,i+\frac{1}{2},j,k} \tag{7.4}$$

added after the curl terms have been assembled. The E_y and E_z first-half systems are modified in exactly the same way, and the second half-step is analogous with $S^{n+\frac{3}{4}}_{E_x,i+\frac{1}{2},j,k}/(2C_{b,i+\frac{1}{2},j,k})$ on the normalized right-hand side.

Consequently a hard electric source (flag $=1$), which would require constraining the unknown E itself rather than adding to the right-hand side, is rejected in the ADI electric path.

7.3 Magnetic Sources After Each Half-Step

After the first ADI half-step has produced $\mathbf{E}^{n+\frac{1}{2}}$ and $\mathbf{B}^{n+\frac{1}{2}}$, and again after the second half-step has produced $\mathbf{E}^{n+1}$ and $\mathbf{B}^{n+1}$, an external magnetic excitation is applied directly to the magnetic field. Write $S^{n}_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}$ for the value assigned to the B_x face at $\left(i,j+\frac{1}{2},k+\frac{1}{2}\right)$ and time level n. With the half-step flag set, a soft magnetic source (flag $=2$) contributes

$$B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow B_{x,i,j+\frac{1}{2},k+\frac{1}{2}}+\tfrac{1}{2}\,S^{n}_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}, \tag{7.5}$$

while a hard magnetic source (flag $=1$) overwrites

$$B_{x,i,j+\frac{1}{2},k+\frac{1}{2}} \leftarrow S^{n}_{B_x,i,j+\frac{1}{2},k+\frac{1}{2}}. \tag{7.6}$$

(The factor $\frac{1}{2}$ again splits a full-step soft increment across the two half-steps.) The same rules apply to B_y and B_z at their staggered faces. Unlike the electric soft source, this magnetic update is an explicit post-processing of the field after the ADI magnetic stage; it does not enter the tridiagonal electric systems.

7.4 Summary of the ADI Excitation Path

Over one full step Δt, the macroscopic ADI scheme therefore proceeds as follows.

1. Form and solve the first-half electric systems with soft electric sources $S_E^{n+\frac{1}{4}}$ added as $\frac{1}{2}S_E^{n+\frac{1}{4}}$ (equivalently $\frac{1}{2}S_E^{n+\frac{1}{4}}/C_b$ on the normalized RHS).
2. Update $\mathbf{B}$ to $n+\frac{1}{2}$ from the ADI magnetic formulae.
3. Apply magnetic excitation to $\mathbf{B}^{n+\frac{1}{2}}$ (soft: add $\frac{1}{2}S_B^n$; hard: set equal to S_B^n).
4. Form and solve the second-half electric systems with soft electric sources $S_E^{n+\frac{3}{4}}$.
5. Update $\mathbf{B}$ to $n+1$, then apply magnetic excitation to $\mathbf{B}^{n+1}$.

Hard electric sources are not part of this path. Soft electric sources never overwrite E; they only augment the Ampère right-hand side of each half-step, consistently with the half-step coefficients C_a and C_b of (3.1).

8 Lumped Inductance

A lumped inductor is introduced as an additional current density $\mathbf{J}_{\mathrm{L}}$ in Ampère's law,

$$\nabla \times \mathbf{H} = \varepsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E} + \mathbf{J}_{\mathrm{L}} \tag{8.1a}$$

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}. \tag{8.1b}$$

On each Yee edge that hosts an inductor, the constitutive relation $V = L\, dI/dt$ for an ideal lumped inductor is written in field form. For an x-directed edge of length Δx and cross-sectional area $\Delta y\, \Delta z$, with lumped inductance L_x assigned to that edge (total inductance, not inductance per unit length),

$$V = E_x \Delta x, \qquad I = J_{\mathrm{L},x}\, \Delta y\, \Delta z, \tag{8.2}$$

so

$$E_x \Delta x = L_x\, \Delta y\, \Delta z\, \frac{\partial J_{\mathrm{L},x}}{\partial t}, \tag{8.3}$$

or equivalently

$$\frac{\partial J_{\mathrm{L},x}}{\partial t} = \kappa_x E_x, \qquad \kappa_x = \frac{\Delta x}{\Delta y\, \Delta z\, L_x}. \tag{8.4}$$

The y and z components are analogous, with

$$\kappa_y = \frac{\Delta y}{\Delta x\, \Delta z\, L_y}, \qquad \kappa_z = \frac{\Delta z}{\Delta x\, \Delta y\, L_z}. \tag{8.5}$$

Away from inductor edges one sets $\kappa = 0$ (equivalently $L \to \infty$), so $\mathbf{J}_{\mathrm{L}}$ is unchanged there and need not be stored or evolved; outside the inductor region, $\mathbf{J}_{\mathrm{L}}$ should normally be initialized to zero. The Maxwell–Faraday equation (8.1b) is unmodified, and therefore the magnetic ADI updates of Section 3 are unchanged. Only the electric update and the inductor ODE are coupled. For FDTD/ADI with lumped R, L, and C, one may treat the element explicitly, semi-implicitly, or fully implicitly. An *explicit* update of J_{L} from a known E (as in the leapfrog macroscopic FDTD inductor) reintroduces a stability restriction set by the local LC scale $\omega_L \sim \sqrt{\kappa/\varepsilon}$, which becomes severe at the large CFLs that motivate ADI. For the isolated lossless field–inductor subsystem, the centered trapezoidal discretization below has amplification factors of unit modulus for every Δt and therefore introduces no lumped-inductor stability restriction; when incorporated consistently into an energy-stable ADI discretization, it preserves the unconditional stability property of the underlying ADI method. A fully one-sided implicit inductor update can also remove the inductor timestep restriction, but introduces numerical dissipation that can be interpreted as an artificial resistive loss. We therefore adopt the centered trapezoidal coupling.

8.1 Centered Trapezoidal Inductor Coupling

Over a half step of size $\Delta t/2$, discretize (8.4) by the trapezoidal rule,

$$\frac{J_{\mathrm{L},x}^{n+\frac{1}{2}} - J_{\mathrm{L},x}^{n}}{\Delta t/2} = \kappa_x\, \frac{E_x^{n+\frac{1}{2}} + E_x^{n}}{2}, \tag{8.6}$$

which rearranges to

$$J_{\mathrm{L},x}^{n+\frac{1}{2}} = J_{\mathrm{L},x}^{n} + \frac{\Delta t}{4}\, \kappa_x \big(E_x^{n+\frac{1}{2}} + E_x^{n}\big). \tag{8.7}$$

For a centered, nondissipative coupling, Ampère's law must contain the averaged current $(J_{\mathrm{L},x}^{n+\frac{1}{2}} + J_{\mathrm{L},x}^{n})/2$, not the fully advanced value $J_{\mathrm{L},x}^{n+\frac{1}{2}}$ alone. Using the same centered conductivity average as in Section 3,

$$\varepsilon \frac{E_x^{n+\frac{1}{2}} - E_x^{n}}{\Delta t/2} + \sigma \frac{E_x^{n+\frac{1}{2}} + E_x^{n}}{2} + \frac{J_{\mathrm{L},x}^{n+\frac{1}{2}} + J_{\mathrm{L},x}^{n}}{2} = [H_1], \tag{8.8}$$

where $[H_1]$ is unchanged from Section 3. From (8.7),

$$\frac{J_{\mathrm{L},x}^{n+\frac{1}{2}} + J_{\mathrm{L},x}^{n}}{2} = J_{\mathrm{L},x}^{n} + \frac{\Delta t}{8}\, \kappa_x \big(E_x^{n+\frac{1}{2}} + E_x^{n}\big). \tag{8.9}$$

Substituting into (8.8) and collecting unknowns gives

$$\left(\frac{2\varepsilon}{\Delta t}+\frac{\sigma}{2}+\frac{\Delta t}{8}\kappa_x\right)E_x^{n+\frac{1}{2}}=\left(\frac{2\varepsilon}{\Delta t}-\frac{\sigma}{2}-\frac{\Delta t}{8}\kappa_x\right)E_x^n+[H_1]-J_{\mathrm{L},x}^n. \tag{8.10}$$

Define the inductor-modified half-step coefficients

$$C_a^L=\frac{\dfrac{2\varepsilon}{\Delta t}-\dfrac{\sigma}{2}-\dfrac{\Delta t}{8}\kappa}{\dfrac{2\varepsilon}{\Delta t}+\dfrac{\sigma}{2}+\dfrac{\Delta t}{8}\kappa},\qquad C_b^L=\frac{1}{\dfrac{2\varepsilon}{\Delta t}+\dfrac{\sigma}{2}+\dfrac{\Delta t}{8}\kappa}. \tag{8.11}$$

When $\kappa=0$ these reduce to (3.1). The electric update is then

$$E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}=C^L_{a,i+\frac{1}{2},j,k}E^n_{x,i+\frac{1}{2},j,k}+C^L_{b,i+\frac{1}{2},j,k}\Big([H_1]-J^n_{\mathrm{L},x,i+\frac{1}{2},j,k}\Big). \tag{8.12}$$

Thus, the inductance modifies the local diagonal electric coefficient through κ in C_a^L and C_b^L, while the previously known current J_{L}^n appears as a known right-hand-side term. After $E_x^{n+\frac{1}{2}}$ is obtained, $J_{\mathrm{L},x}$ is updated by (8.7).
On an inductor edge, where $\kappa>0$, and with $\sigma=0$ and no curl contribution, define the local electromagnetic–inductor energy by

$$\mathcal{E}=\frac{\varepsilon}{2}E^2+\frac{1}{2\kappa}J_{\mathrm{L}}^2. \tag{8.13}$$

This energy is exactly conserved by the centered pair

$$\varepsilon\frac{E^{n+\frac{1}{2}}-E^n}{\Delta t/2}+\frac{J_{\mathrm{L}}^{n+\frac{1}{2}}+J_{\mathrm{L}}^n}{2}=0,\qquad \frac{J_{\mathrm{L}}^{n+\frac{1}{2}}-J_{\mathrm{L}}^n}{\Delta t/2}=\kappa\frac{E^{n+\frac{1}{2}}+E^n}{2}, \tag{8.14}$$

so $\mathcal{E}^{n+\frac{1}{2}}-\mathcal{E}^n=0$. By contrast, replacing the averaged current in Ampère by the fully advanced value $J_{\mathrm{L}}^{n+\frac{1}{2}}$ yields

$$\mathcal{E}^{n+\frac{1}{2}}-\mathcal{E}^n=-\frac{1}{2\kappa}\big(J_{\mathrm{L}}^{n+\frac{1}{2}}-J_{\mathrm{L}}^n\big)^2\le 0, \tag{8.15}$$

and is therefore numerically dissipative even though the inductor ODE itself is integrated by the trapezoidal rule.

8.2 Effect on the ADI Tridiagonal System

The magnetic update for $H_z^{n+\frac{1}{2}}$ is identical to Section 3. Substituting it into (8.12) yields the same stencil structure as (3.9), but with $C_a\to C_a^L$ and $C_b\to C_b^L$:

$$\begin{aligned}-\alpha_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j-1,k}+\beta^L_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j,k}-\gamma_{x1}E^{n+\frac{1}{2}}_{x,i+\frac{1}{2},j+1,k}=\;&p^L_{x1}E^n_{x,i+\frac{1}{2},j,k}\\&+\frac{1}{\Delta y}\Big(H^n_{z,i+\frac{1}{2},j+\frac{1}{2},k}-H^n_{z,i+\frac{1}{2},j-\frac{1}{2},k}\Big)\\&-\frac{1}{\Delta z}\Big(H^n_{y,i+\frac{1}{2},j,k+\frac{1}{2}}-H^n_{y,i+\frac{1}{2},j,k-\frac{1}{2}}\Big)\\&+\frac{q_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j-\frac{1}{2},k}-E^n_{y,i,j-\frac{1}{2},k}\Big)\\&-\frac{r_{x1}}{\Delta x\Delta y}\Big(E^n_{y,i+1,j+\frac{1}{2},k}-E^n_{y,i,j+\frac{1}{2},k}\Big)\\&-J^n_{\mathrm{L},x,i+\frac{1}{2},j,k},\end{aligned} \tag{8.16}$$

where

$$\alpha_{x1} = \frac{D_{b,i+\frac{1}{2},j-\frac{1}{2},k}}{\Delta y^2}, \tag{8.17}$$

$$\gamma_{x1} = \frac{D_{b,i+\frac{1}{2},j+\frac{1}{2},k}}{\Delta y^2}, \tag{8.18}$$

$$\beta^L_{x1} = \frac{1}{C^L_{b,i+\frac{1}{2},j,k}} + \alpha_{x1} + \gamma_{x1}, \tag{8.19}$$

$$p^L_{x1} = \frac{C^L_{a,i+\frac{1}{2},j,k}}{C^L_{b,i+\frac{1}{2},j,k}}, \tag{8.20}$$

$$q_{x1} = D_{b,i+\frac{1}{2},j-\frac{1}{2},k}, \tag{8.21}$$

$$r_{x1} = D_{b,i+\frac{1}{2},j+\frac{1}{2},k}. \tag{8.22}$$

The off-diagonal coefficients α_{x1} and γ_{x1} are unchanged (they come only from the implicit curl of H). The diagonal β^L_{x1} and the factor p^L_{x1} feel L_x through C^L_b and C^L_a. The E_y and E_z first-half systems are modified in exactly the same way. The second half-step is analogous: the current update is

$$J^{n+1}_{\mathrm{L},x} = J^{n+\frac{1}{2}}_{\mathrm{L},x} + \frac{\Delta t}{4}\,\kappa_x\big(E^{n+1}_x + E^{n+\frac{1}{2}}_x\big), \tag{8.23}$$

while the corresponding Ampère stage uses the centered current $(J^{n+1}_{\mathrm{L},x} + J^{n+\frac{1}{2}}_{\mathrm{L},x})/2$, which after elimination produces the same $\Delta t\,\kappa/8$ modification of the electric coefficients and the known right-hand-side term $-J^{n+\frac{1}{2}}_{\mathrm{L},x}$.

8.3 Stability Remarks

Eliminating J_L locally produces an oscillator of frequency $\omega_L = \sqrt{\kappa/\varepsilon}$. For the isolated lossless field–inductor subsystem, the centered trapezoidal discretization has amplification factors of unit modulus for every Δt and therefore introduces no lumped-inductor stability restriction. Unit-modulus amplification of the local oscillator alone is not, by itself, a complete proof of unconditional stability of the full spatial ADI splitting; that property follows only when the inductor coupling is embedded consistently in an energy-stable ADI discretization. By contrast, replacing (8.7) by the explicit rule $J^{n+\frac{1}{2}}_\mathrm{L} = J^n_\mathrm{L} + (\Delta t/2)\kappa E^n$ and inserting that known current only as a right-hand-side source leaves $\omega_L\Delta t$ constrained, defeating the purpose of ADI at large CFL.

8.4 Remark on the Present Implementation

In the current Artemis macroscopic ADI push, soft electric sources are added to the ADI right-hand side, but the lumped-inductor current $\mathbf{J}_\mathrm{L}$ is not. The inductor object still advances $\mathbf{J}_\mathrm{L}$ explicitly when enabled, yet that current is unused by `MacroscopicEvolveADI`. The modification needed for a stable ADI inductor is therefore: (i) replace C_a, C_b by C^L_a, C^L_b on inductor edges; (ii) add $-C^L_b J^{\mathrm{old}}_\mathrm{L}$ to the field-update form, or equivalently add $-J^{\mathrm{old}}_\mathrm{L}$ to the normalized tridiagonal right-hand side; and (iii) update J_L with the trapezoidal rule (8.7) after each electric solve.